

2U MATHEMATICS ASSESSMENT

2 periods - Week 8, Term 4 - 1993.

Answer all questions.

Show all working.

Non-programmable calculators may be used.

Question 1 (16 marks)

(a) Evaluate $2\pi\sqrt{\frac{c}{f}}$ when $c = 6.3$

and $f = 7.9$, to 2 significant figures.

(b) The line $3x - py = 5$ passes through the point $(3,2)$, what is the value of p ?

(c) Simplify $3\sqrt{27} + \sqrt{12}$

(d) Express $\frac{3}{2 + \sqrt{5}}$ as a single fraction with a rational denominator.

(e) Find the value of $\frac{3.67^2 + 10.06^{-3}}{-7.8}$ correct to 2 decimal places.

(f) Express $0.2\dot{i}$ as a rational number.

(g) Expand and simplify $(3x-2)^3$

(h) Express $6ab + a^2 - 2b^2 - 2ab$ in its simplest form.

(i) If $a = -2$, $b = 8$, $c = -1$

evaluate $\frac{(a+c)^2}{b^2}$

QUESTION 2. (12 marks)

(i) Solve (a) $5(a-3) = 2(a+6)$

(b) $|5x+2| < 12$

(ii) Find the exact roots of the quadratic equation $5x^2 - 3x - 1 = 0$

(iii) Two numbers x and y are connected by the equations $x-y = 2$ and $xy = 5$. Find the values of x and y .

(iv) Find $\lim_{x \rightarrow 2} \frac{x^3 - 8}{2x - 4}$

(v) If $\sin \theta = \frac{2}{\sqrt{29}}$ and θ is obtuse

find the exact values of

(a) $\tan \theta$, (b) $\cos \theta$

QUESTION 3. (12 marks)(a) Find the derivatives of the following functions of x .

(i) $5x^3 - 4x^2 + \sqrt{x}$

(ii) $x^2 + \frac{3}{x^2}$

(iii) $(x-1)(x^2+x+1)$

(b) (i) Sketch $y = \sqrt{25 - x^2}$

(ii) State the domain and range of this function.

(c) What is the exact value of $\sec 135^\circ$?

(d) Show that $\frac{x^2}{9} + \frac{y^2}{4} = 1$

if $x = 3 \cos A$ and $y = 2 \sin A$

QUESTION 4 (12 marks)

- (a) Sketch the function defined by the rule

$$f(x) = \begin{cases} 2x - 1 & \text{if } x \geq 1 \\ 2 & \text{if } x < 1 \end{cases}$$

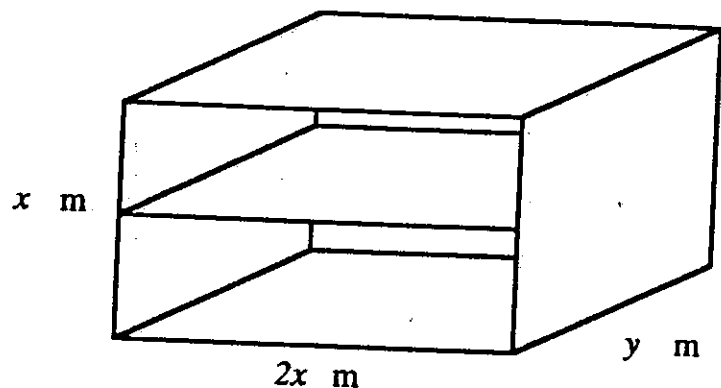
- (b) Find the equation of the tangent to the curve $y = 6x - 2x^2$ at the point where $x = 2$.
What is the equation of the normal at this point?

- (c) For the curve $y = x^3 - 12x$

- (i) What are the intercepts on the x and y axes?
(ii) Find the stationary points and determine their nature.
(iii) Thus sketch the curve showing all relevant details.

QUESTION 5 (12 marks)

- (a)



A rectangular storage unit, closed on all sides except the front, is to be constructed, with a shelf, as shown. It is to be built with 24m^2 of pineboard (disregard thickness)

- (i) Show that $2x^2 + 8xy = 24$
(ii) Form a function, $V(x)$, that measures the volume of the storage unit.
(iii) Determine the maximum volume the storage unit can have.

(b) PQRS is a rhombus in which the side lengths are 4 cm., the angle PQR is 60° and the diagonals intersect at M. The angle PMQ is denoted by ϕ .

- (i) Draw a diagram to represent this information.
(ii) Explain why ϕ is a right angle
(iii) Calculate the exact length of the diagonal PR.
(iv) Calculate the exact length of the other diagonal SQ.

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QUESTION 6 (12 marks)

(a) Differentiate the following with respect to x .

(i) $\frac{1}{4\sqrt{x}}$

(ii) $\frac{5x-3}{2x^3+1}$

(iii) $(3x - 1)^6$

(b) A set of 11 concentric circles is such that the circumference of each is 2π cm. more than the one immediately inside it. The innermost circle has a diameter of 20 cm.

Find (i) the circumference of the outermost circle.

(ii) the diameter of a circle whose circumference is equal to the total of the circumferences of the 11 circles.

YEAR 11 TERM 4 MATHS ASSESSMENT SOLUTIONS.

Question 1:

(a) $5 \cdot 6$ (2.s.f.)

(b) $3x - py = 5$

$9 - 2p = 5$

$p = 2$

(c) $3\sqrt{27} + \sqrt{12}$

$= 3 \cdot 3\sqrt{3} + 2\sqrt{3}$

$= 11\sqrt{3}$

(d) $\frac{3}{2+\sqrt{5}} \times \frac{2-\sqrt{5}}{2-\sqrt{5}}$

$= \frac{6 - 3\sqrt{5}}{4 - 5}$

$= 3\sqrt{5} - 6$

(e) -1.73 (2d.p.)

(f) $x = 0.2\bar{1} = 0.2111\dots$

$10x = 2.111\dots$

$100x = 21.111\dots$

$90x = 19$

$x = \frac{19}{90}$

(g) $(9x^2 - 12x + 4)(3x - 2)$

$= 27x^3 - 36x^2 + 12x - 18x^2 + 24x - 8$

$= 27x^3 - 54x^2 + 36x - 8$

(h) $6ab + a^2 - 2b^2 - 2ab$

~~$= a(6b + a) - 2b(b + a)$~~

$= 4ab + a^2 - 2b^2$

$= a^2 + 4ab - 2b^2$

(i) $\frac{9}{64}$

Question 2

(1) (a) $5(a-3) = 2(a+6)$

$5a - 15 = 2a + 12$

$3a = 27$

$a = 9$

(b) $|5x+2| < 12$

$5x+2 < 12$ or $5x+2 > -12$

$5x < 10$

$5x > -14$

$x < 2$

$x > -\frac{14}{5}$

(2)

$x > -2\frac{4}{5}$

$-2\frac{4}{5} < x < 2$

(11) $5x^2 - 3x - 1 = 0$

$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

(2) $= \frac{3 \pm \sqrt{9 + 4 \times 5 \times 1}}{10} = \frac{3 \pm \sqrt{29}}{10}$

(111) $x - y = 2$ - (1) $\Rightarrow y = x - 2$

$xy = 5$ - (2)

$\therefore x(x-2) = 5$

$x^2 - 2x - 5 = 0$

$x = \frac{2 \pm \sqrt{4 + 4 \times 1 \times 5}}{2} = \frac{2 \pm \sqrt{4 + 20}}{2}$

$= \frac{2 \pm 2\sqrt{6}}{2} = 1 \pm \sqrt{6}$

(1) $x = 1 + \sqrt{6}$ or $1 - \sqrt{6}$

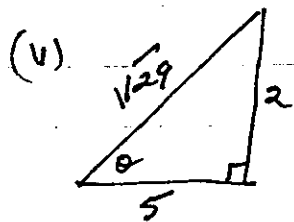
(1) $y = -1 + \sqrt{6}$ or $-1 - \sqrt{6}$

(using $y = x - 2$)

(1V) $\lim_{x \rightarrow 2} \frac{x^3 - 8}{x^2 - 4} = \lim_{x \rightarrow 2} \frac{(x-2)(x^2 + 2x + 4)}{2(x-2)}$

$= 6$

(2)



(a) $\tan \theta = -\frac{2}{5}$ ①

(b) $\cos \theta = \frac{-5}{\sqrt{29}}$ ①

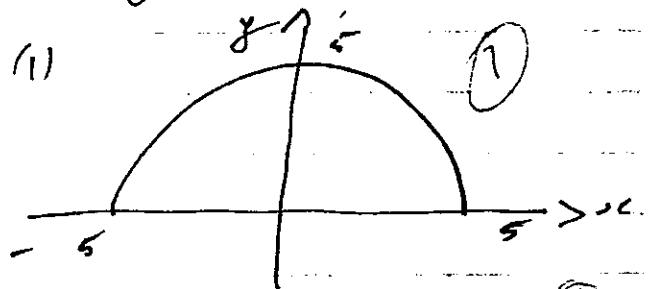
Question 3

(a) (i) $y = 5x^3 - 4x^2 + \sqrt{x}$
 $= 5x^3 - 4x^2 + x^{\frac{1}{2}}$
 $\frac{dy}{dx} = 15x^2 - 8x + \frac{1}{2}x^{-\frac{1}{2}}$
 $= 15x^2 - 8x + \frac{1}{2\sqrt{x}}$ ①

(ii) $y = x^2 + \frac{3}{x^2}$
 $= x^2 + 3x^{-2}$
 $\frac{dy}{dx} = 2x - 6x^{-3}$
 $= 2x - \frac{6}{x^3}$ ②

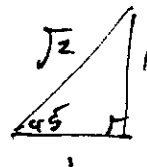
(iii) $y = (x-1)(x^2+x+1)$
 $\Rightarrow u \cdot v$
 $u = x-1 \quad v = x^2+x+1$
 $u' = 1 \quad v' = 2x+1$
 $\frac{dy}{dx} = v u' + u v'$
 $= (x^2+x+1) \cdot 1 + (x-1)(2x+1)$
 $= x^2+x+1 + 2x^2+x-2x-1$
 $= 3x^2$ ②

(b) $y = \sqrt{25-x^2}$



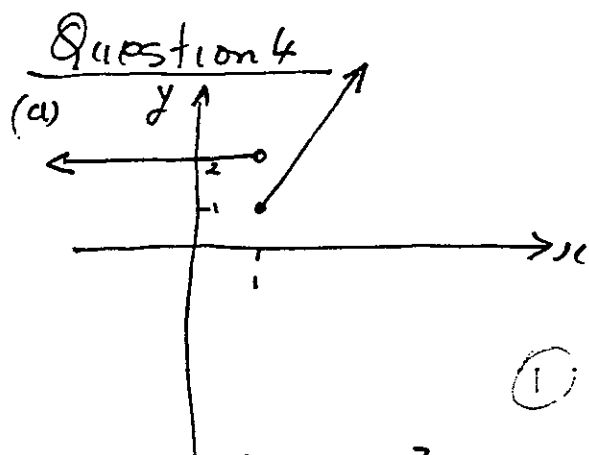
(ii) $D: -5 \leq x \leq 5$ ①
 $R: 0 \leq y \leq 5$ ①

(c) $\sec 135 = \sec (180-45)$
 $= -\sec 45$
 $= -\frac{1}{\cos 45}$
 $= -\sqrt{2}$ ②



(d) $\frac{x^2}{9} + \frac{y^2}{4} = 1$

LHS: $\frac{(3\cos A)^2}{9} + \frac{(2\sin A)^2}{4}$
 $= \frac{9\cos^2 A}{9} + \frac{4\sin^2 A}{4}$ ①
 $= \cos^2 A + \sin^2 A$
 $= 1$
 $= \text{RHS}$ ①



(b) $y = 6x - 2x^2$, $x=2$.
 Tangent: $y' = 6 - 4x$
 $x=2$, $m = -2$.
 $y = 12 - 8 = 4$.

$y - y_1 = m(x - x_1)$
 $y - 4 = -2(x - 2)$
 $y - 4 = -2x + 4$
 $2x + y - 8 = 0$

NORMAL $y - 4 = \frac{1}{2}(x - 2)$.
 $2y - 8 = x - 2$
 $x - 2y + 6 = 0$

(c) $y = x^3 - 12x$.
 (i) $y = x(x^2 - 12)$
 $= x(x - \sqrt{12})(x + \sqrt{12})$
 intercepts $x = 0, \sqrt{3}, -\sqrt{3}$

(ii) $\frac{dy}{dx} = 3x^2 - 12$
 S.P. when $\frac{dy}{dx} = 0$

$3x^2 - 12 = 0$
 $x^2 - 4 = 0$

$(x-2)(x+2) = 0$.
 S.P. at $x = 2, -2$.
 $y = -16, 16$

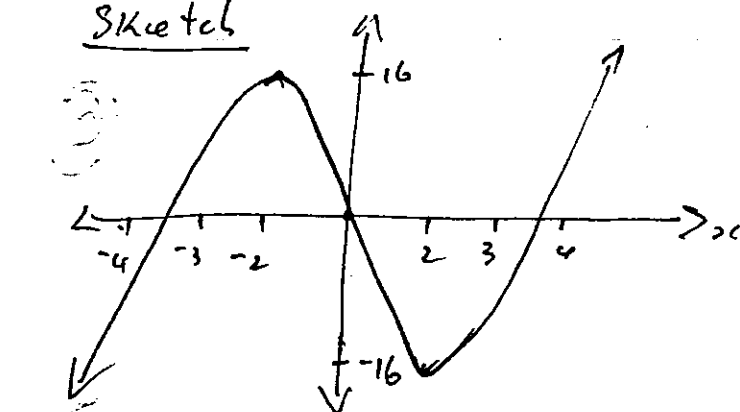
$(2, -16)$ $(-2, 16)$

TEST

x	2^-	2	2^+	
$\frac{dy}{dx}$	-	0	+	$(2, -16)$ → a Min. T.P.

x	-2^-	-2	-2^+	
$\frac{dy}{dx}$	+	0	-	$(-2, 16)$ → a Max. T.P.

Sketch



Question 5

(i) Total Surface Area.

$= 2xy + 3 \times 2xy + 2x^2$
 $= 2x^2 + 8xy$

$\therefore 2x^2 + 8xy = 24$ — (1)

(ii) $V = 2x^2y$

(iii) $V = 2x^2 \left(\frac{24 - 2x^2}{8x} \right)$ sub in (1)

$= 2x^2 \left(\frac{12 - x^2}{4x} \right)$

$= x^2 \left(\frac{12 - x^2}{2} \right)$

$= \frac{12x^2 - x^4}{2} = 6x^2 - \frac{1}{2}x^3$

$\frac{dV}{dx} = 12 - \frac{3}{2}x^2$

$$\frac{dv}{dx} = 0$$

$$6 = \frac{3}{2}x^2$$

$$12 = 3x^2$$

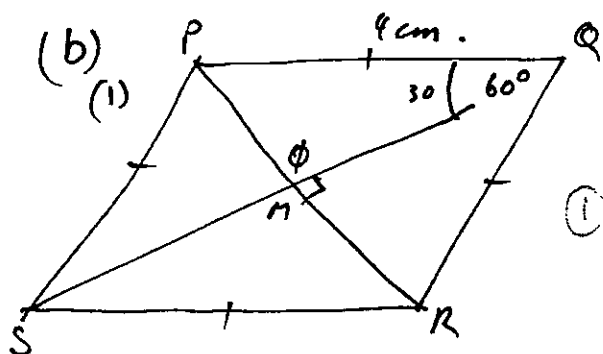
$$4 = x^2$$

$$x = \pm 2$$

$$x = 2 \quad \begin{array}{c|c|c|c} x & 2 & 2 & 2^3 \\ \hline dy/dx & + & 0 & - \end{array}$$

Max at $x = 2$ ①

$$\begin{aligned} \therefore V_{\max} &= 6 + 2 - \frac{1}{2} \cdot 2^3 \quad ① \\ &= 12 - 4 \\ &= 8 \text{ m}^3 \quad ① \end{aligned}$$



(ii) Diagonals of a rhombus bisect at right angles ①

$$\begin{aligned} \text{(iii)} \quad \sin 30 &= \frac{PM}{4} \\ PM &= 4 \sin 30 \\ &= 4 \times \frac{1}{2} \\ &= 2 \end{aligned}$$

$$\begin{aligned} PR &= 2 PM \\ &= 4 \text{ units} \quad ① \end{aligned}$$

$$\begin{aligned} \cos 30 &= QM/4 \\ QM &= 4 \cos 30 \\ &= 2\sqrt{3} \quad \therefore QS = 4\sqrt{3} \end{aligned}$$

Questions

$$\begin{aligned} \text{(1)} \quad y &= \frac{1}{4\sqrt{x}} = \frac{1}{4} \times x^{-\frac{1}{2}} \\ \frac{dy}{dx} &= -\frac{1}{8} x^{-\frac{3}{2}} \end{aligned}$$

$$\text{(2)} \quad = -\frac{1}{8} \times \frac{1}{\sqrt{x^3}} = -\frac{1}{8\sqrt{x^3}} \quad \#$$

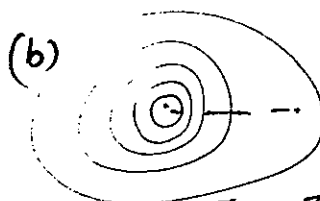
$$\begin{aligned} \text{(ii)} \quad y &= \frac{5x-3}{2x^3+1} \Rightarrow \frac{u}{v} \\ \frac{dy}{dx} &= \frac{vu' - uv'}{v^2} \end{aligned}$$

$$\begin{aligned} u &= 5x-3 \\ u' &= 5 \\ v &= 2x^3+1 \\ v' &= 6x^2 \\ v^2 &= (2x^3+1)^2 \end{aligned}$$

$$\begin{aligned} &= \frac{(2x^3+1)5 - (5x-3)(6x^2)}{(2x^3+1)^2} \\ &= \frac{10x^3+5 - 30x^3+18x^2}{(2x^3+1)^2} \end{aligned}$$

$$\text{(2)} = \frac{5 + 18x^2 - 20x^3}{(2x^3+1)^2} \quad \#$$

$$\begin{aligned} \text{(iii)} \quad y &= (3x-1)^6 \\ \frac{dy}{dx} &= 6(3x-1)^5 \cdot 3 \\ &= 18(3x-1)^5 \quad \# \end{aligned}$$



$$\begin{aligned} c &= 2\pi r \\ &= \pi d \end{aligned}$$

$$\begin{aligned} T_1 \quad T_2 \quad T_3 \quad T_4 \quad T_5 \quad \dots \quad T_n \\ 20 \quad 20+12\pi \quad 20+4\pi \quad \dots \quad 20+20\pi \end{aligned}$$

$$\text{(1)} \quad 20\pi + 20\pi = 40\pi \quad ②$$

$$\begin{aligned} \text{(ii)} \quad S_n &= \frac{n}{2}(a+l) \\ &= \frac{n}{2}(40 + 20\pi) \\ &= \frac{n}{2}(40 + 20\pi) = 11(20 + 10\pi) \\ &= 11 \times 30\pi = 330\pi \\ \pi d &= 330\pi \quad \text{diameter} = \frac{330\pi}{\pi} = 330 \text{ cm} \end{aligned}$$